Transport Starter Data Kit: Historical socio-transport data for South Africa

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Abstract

Data on transport activity is an important element for the development of national transport decarbonisation strategies. By having freight and passenger transport information, the impacts on vehicle and fuel consumption changes from replacing internal combustion engine vehicles with electric vehicles can be calculated. The development of a national decarbonisation strategy requires significant efforts. However, access to data is often a barrier to starting transport system modelling in developing countries, thereby causing delays. This article provides data that can be used to support a model for South Africa, which may act as a starting point for further model development and scenario analysis. The data are collected entirely from publicly available and accessible sources, focusing on national reports, statistical yearbooks, and academia.

Keywords

U4RIA

Transport data

Transport modelling

MAED

South Africa

Specifications Table

Subject	Transport
Specific subject area	Transport Data
Type of data	Tables Graphs
How data were acquired	Literature survey (databases and reports from international organisations; journal articles)
Data format	Raw and analysed
Parameters for data collection	Data collected based on inputs required to create an energy system model for South Africa
Description of data collection	Data were collected from the websites, annual reports and databases of international organisations, as well as from academic articles and existing modelling databases.
Data source location	Not applicable
Data accessibility	With the article and in a repository. Repository name: Zenodo. Direct URL to data: https://doi.org/10.5281/zenodo.6540022

Value of the Data

- The data can be used to develop national transport demand models to inform national investment outlooks and decarbonisation strategies.
- The data are useful for country analysts, policy makers, and the broader scientific community, as a zero-order starting point for model development.
- This data could be used to examine a range of possible transport pathways, in addition to the examples given in this study, to provide further insights into the evolution of South Africa's transport system.
- The data can be used both for conducting an analysis of transport activity and emissions, but also for capacity building activities.
- The data can be used as a call to action in addressing transport data gaps and establishing parameters for data collection to improve the consistency of transport-climate research in these countries.

1. Data Description

The data provided in this paper can be used to support the development of a transport model for South Africa. The data provided were collected from publicly available sources, including statistical yearbooks, transport ministry reports, statistics from national authorities and affiliated research institutions, academia, and journal articles. Global datasets (primarily from the World Bank) were only consulted if severe data gaps existed. The dataset includes parameters on passenger and freight transport activity, disaggregated by transport mode (road, rail, aviation, etc.) and geographic scale (inter-city or inner-city), if available. The dataset also covers the size of the vehicle fleet, disaggregated by vehicle types. The data coverage and subtypes vary among the parameters. The overall ambition is to include the most recent available year(s).

Item	*Description of Content*
Figure 1	A graph showing total population (million people), as well as the share of urban and rural population in South Africa.
Figure 2	A graph showing total GDP (million USD in 2015), as well as the share of the different sectors contributing to GDP in South Africa: agriculture, construction, mining, manufacturing, service, and energy.
Table 1	A table showing passenger transport activity in South Africa for the most recent year data was available. The data are curated from national statistics agencies or other government-affiliated agencies.
Table 2	An additional table showing passenger transport activity in South Africa based on UN DESA Statistics Division data (see explanation below). The data feature information for 2018.
Tables 3 to 5	Tables showing freight transport activity in South Africa for the most recent year data was available.
Table 6	An additional table showing freight transport activity in South Africa based on UN DESA Statistics Division data (see explanation below). The data feature information for 2018.
Table 7	Tables showing vehicle fleet data in South Africa for the most recent year data were available.

For the parameters on passenger and freight transport activity, an additional dataset was included in Table 2 and Table 6. The UN DESA Statistics Division modelled passenger activity and freight activity for every country in support of SDG Indicator 9.1.2[1]. Passenger activity data provide information for road, rail, and air transport. Freight data cover road, rail and inland water, and aviation. The passenger-km and tonnes-km data originate from the Open SDG Data Hub. In this dataset, only the data for International Transport Forum (ITF) (representing mostly OECD countries) and UNECE countries (mostly European countries) are based on national reporting. For non-ITF/UNECE countries, the data are estimated using the ITF model, which uses several covariates such as GDP, population, and transport network coverage. A description of the model can be found in the ITF Transport Outlook 2017.

1.1 Population

Population data including total population, population growth, and split by rural or urban was gathered from The World Bank Open Data platform[2]. Figure 1 displays the total population disaggregated by urban and rural in South Africa.

[1] Freight: https://www.sdg.org/datasets/undesa::indicator-9-1-2-freight-volume-by-mode-of-transport-tonne-kilometres/about ;
Passenger: https://www.sdg.org/datasets/undesa::indicator-9-1-2-passenger-volume-passenger-kilometres-by-mode-of-transport/about

[2] https://data.worldbank.org/

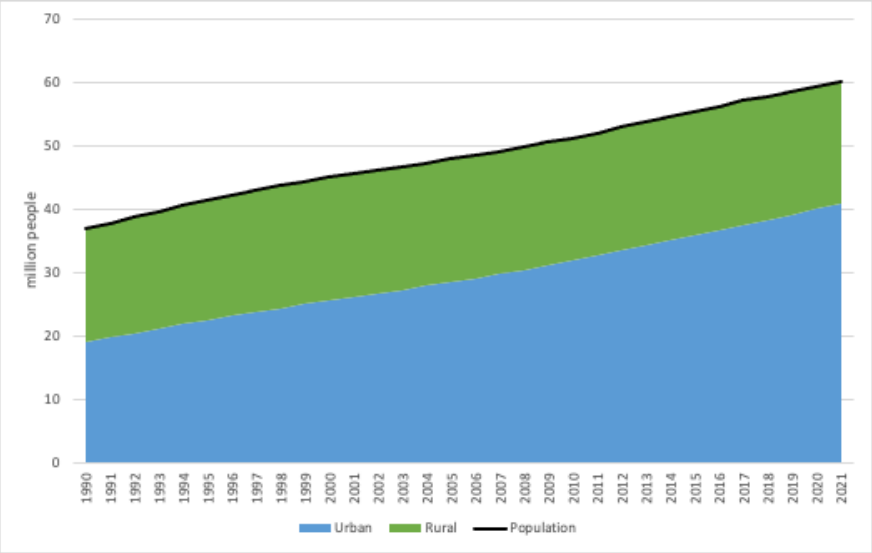


Figure 1: Total population (million people) disaggregated by urban and rural in South Africa

1.2 Gross domestic product (GDP)

GDP data including total GDP, GDP growth, and GDP share by sector (agriculture, manufacturing, service) was collected from The World Bank Open Data platform[2]. Where data was not available, data processing was done. Figure 2 shows the total GDP, as well as the share by sector, in South Africa.

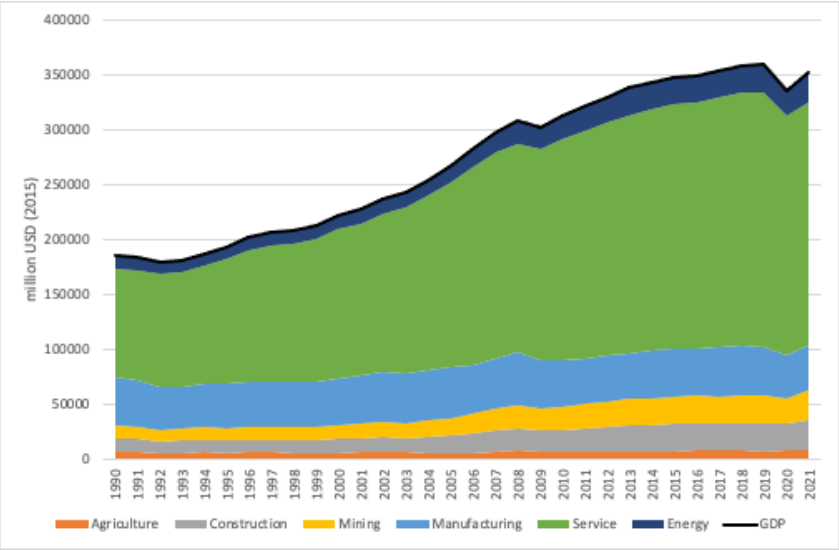


Figure 2: Total GDP (million USD in 2015) disaggregated by share in South Africa

1.3 Passenger transport activity

Information on passenger transport activity in South Africa is provided for major transport modes, covering buses, mini-bus taxis, aviation, rail, passenger cars, sport utility vehicles (SUVs), and motorcycles (Table 1).

Mini-bus taxis represented 117400 million passenger-km in 2006 the transport mode with the highest passenger activity. Passenger cars record 102500 million passenger-km. The issue with the data is that it is only available for 2006. It was released as part of a bottom-up model by the University of Cape Town.

Table 1: Passenger transport activity (million passenger-km) in South Africa

Mode	2006
Large bus	15700
Mini-bus taxis	117400
Rail	13900
Air	11400
Passenger car	102500
SUV	13600
Motorcycle	2400

Source: Merven, B., Stone, A., Hughes, A. and Cohen, B., 2012, Quantifying the energy needs of the transport sector for South Africa: A bottom-up model. University of Cape Town, Cape Town, https://open.uct.ac.za/handle/11427/16905

The UN DESA modelled data estimated that the passenger activity in South Africa is 262342 million passenger-km for road, 16430 million passenger-km for rail, and 34652 million passenger-km for aviation in 2018 (Table 2). The large majority of passenger activity is conducted through road transport.

Table 2: Modelled passenger transport activity (million passenger-km) in South Africa

Mode	2018
Aviation	34652.33215
Rail	16430.49669
Road	262342.774

Comparing the values in Tables 1 and 2, the modelled estimates by UN DESA do not differ too much for road and rail. Only the estimates for aviation are three times as high as the estimates by the University of Cape Town.

1.4 Freight transport activity

The information on freight transport activity is also retrieved from academia, as the South African government does not seem to release any statistics on this topic. This subsection introduces two different studies. The first study focuses on road vehicles, while the second study provides information on all land transport modes and by geography (metropolitan, corridors, and rural areas).

The bottom-up model by the University of Cape Town presents freight transport activity by light-commercial vehicles, medium-commercial vehicles, and heavy-commercial vehicles (Table 3). The largest share of road freight is transported via heavy-commercial vehicles (120100 million tonnes-km in 2006).

Table 3: Freight transport activity (million tonnes-km) in 2006 in South Africa

Mode	2006
Light-commercial vehicle	14400
Medium-commercial vehicle	9100
Heavy-commercial vehicles	120100

Source: Merven, B., Stone, A., Hughes, A. and Cohen, B., 2012, Quantifying the energy needs of the transport sector for South Africa: A bottom-up model. University of Cape Town, Cape Town, https://open.uct.ac.za/handle/11427/16905

The Stellenbosch University released the Logistics Barometer for South Africa in 2016. It covers information for roads (divided into line-haul and distribution), railways, and pipelines for 2013 and 2014 (Table 4). Rail transport records 139000 million tonnes-km, road (line-haul and distribution combined) records 231000 million tonnes-km, and pipelines 7900 million tonnes-km in 2014.

Table 4: Freight transport activity (million tonnes-km) for land transport in South Africa

Mode	2013	2014
Road line-haul	151000	158000
Road distribution	72000	73000
Rail	130000	139000
Pipeline	7800	7900

Source: Havenga, J.H., Simpson, Z.P., King, D. de Bod, A. and Braun, M, 2016, Logistics Barometer South Africa 2016. Stellenbosch University, http://www.sun.ac.za/english/faculty/economy/logistics/Pages/logisticsbarometer.aspx

The report further divides road and rail freight transport for 2014 into corridor, metropolitan, and rural (Table 5). It shows that road freight is essential for innercity goods transport. Metropolitan road freight represents 77022 million tonnes-km.

Table 5: Freight transport activity (million tonnes-km) in South Africa

Mode	Sub-type	2014
Road	Corridor	103411
Road	Metropolitan	77022
Road	Rural	50560
Rail	Corridor	27489
Rail	Metropolitan	800
Rail	Rural	12640

Source: Havenga, J.H., Simpson, Z.P., King, D. de Bod, A. and Braun, M, 2016, Logistics Barometer South Africa 2016. Stellenbosch University, http://www.sun.ac.za/english/faculty/economy/logistics/Pages/logisticsbarometer.aspx

According to the UN DESA modelled data, freight activity for 2018 in South Africa is 172133 million tonnes-km by road, 129066 million-tonnes km by rail, 716 million tonnes-km by aviation, and 78 million tonnes-km

by inland waterways (Table 6). Freight transport is mostly achieved through road transport but rail transport is also seen as very significant.

Table 6: Modelled freight transport activity (million tonnes-km) in South Africa

Mode	2018
Aviation	716.246208
Inland waterways	78.931003
Rail	129066.1229
Road	172133.2927

Comparing the estimates with results in Tables 3 to 5 shows great divergence. Road freight is lower in the UN DESA database than it is in the studies. The data points for aviation and inland waterways cannot be compared due to missing information. But the overall trend, that both rail and roads are major freight carriers, is correct.

1.5 Vehicle fleets

South Africa has a total of 5743500 road vehicles, as of 2006. 78% of road vehicles are passenger cars. Even though the previous sections show that mini-bus taxis record the largest passenger activity, it only represents less than 5% of the total vehicle fleet. This information is based on the model by the University of Cape Town. Official records were not accessible. The data is only available for 2006, thus it does not allow for analysis of the potential growth and development of the vehicle fleet.

Table 7: Vehicle fleet numbers in South Africa

Mode	2006
Large bus	221000
Mini-bus taxis	269300
Passenger car	4488500
SUV	498400
Motorcycle	266300
Total vehicles	5743500

Source: Merven, B., Stone, A., Hughes, A. and Cohen, B., 2012, Quantifying the energy needs of the transport sector for South Africa: A bottom-up model. University of Cape Town, Cape Town, https://open.uct.ac.za/handle/11427/16905

2. Methodology

The focus is on national data for passenger activity (passenger-km), freight activity (tonnes-km) and modes of transport (number of vehicles). The priority is to collect data released by national governments, government-affiliated organisations, or country-specific studies. The research identifies the most recent available data and any data available from 1990 onwards. The priority was for any data after 2010, because

transport is a very dynamic growth sector and anything before 2010 adds limited value to understanding the current real-world situation.

Desk research is the main data collection approach for the Transport Starter Data Kits. The desk research examined annual yearbooks, transport statistics, country reporting, and any national statistical portals. Websites of the national government, transport ministries, statistical institutes and other related authorities were examined. Only when severe data gaps exist, global datasets are consulted. In some cases, World Bank data[3] on rail passenger and rail freight is included.

Each Transport Data for Starter Data Kit set contains an additional dataset, which is sourced from the United Nations Department of Economic and Social Affairs (UN DESA) Statistics Division. It is included as a secondary priority because this dataset is the result of a modelling exercise and covers every country. The UN DESA modelled passenger activity and freight activity has the purpose to support the Sustainable Development Goal Indicator 9.1.2[4]. The passenger activity provides information for road, rail, and air transport. Freight data covers the road, rail and inland water, and aviation. The passenger-km and tonnes-km data originate from the Open Sustainable Development Goals (SDG) Data Hub. In the UN DESA dataset, only the data for countries participating in the International Transport Forum (ITF) (representing mostly member countries of the Organisation for Economic Co-operation and Development (OECD)) and the United Nations Economic Commission for Europe (UNECE) (mostly European countries) are based on national reporting. For non-ITF/UNECE countries, data are estimated using the ITF model, which uses several covariates such as gross domestic product, population, and transport network coverage. A description of the model can be found in the ITF Transport Outlook 2017[5]. The UN DESA dataset is included in the Transport Data for Starter Data Kits as additional tables to fill in the incomplete picture that most countries present. The UN DESA modelled data is less accurate and it shall only be regarded as offering the wider picture of transport activity in the country.

The collected data have been shared with a group of relevant SLOCAT partners to validate and explore any additional sources. The SLOCAT partners were selected based on their actions to lead projects in the region and their involvement in data-focused knowledge products or projects. The consultation involved ten anonymous organisations.

Desk research is an approach that limits the research to material available on the internet, accessible through search engines and linked to government and statistical institutes' websites. However, this does not pose a major limitation to obtaining data. Nearly every country has functional websites for statistics and transport authorities. In a few cases, websites are not well maintained, resulting in missing or broken hyperlinks to reports. By using services that provide access to archived websites, some of these broken

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[3] Rail passenger data: World Bank, 2022, Railways, passengers carried (million passenger-km), https://data.worldbank.org/indicator/IS.RRS.PASG.KM; rail freight data: World Bank, 2022, Railways, goods transported (million ton-km), https://data.worldbank.org/indicator/IS.RRS.GOOD.MT.K6

[4] UN DESA, 2021, Indicator 9.1.2: Freight volume by mode of transport (tonne kilometres): https://www.sdg.org/datasets/undesa::indicator-9-1-2-freight-volume-by-mode-of-transport-tonne-kilometres/about ;
UN DESA, 2021, Indicator 9.1.2: Passenger volume (passenger kilometres) by mode of transport: https://www.sdg.org/datasets/undesa::indicator-9-1-2-passenger-volume-passenger-kilometres-by-mode-of-transport/about

[5] ITF, 2017, ITF Transport Outlook 2017, https://www.itf-oecd.org/transport-outlook-2017

pages can be retrieved. The collected information has been shared with partners and no additional information has been received.

While over 1,500 languages are spoken across Sub-Saharan Africa, government datasets are generally published in a smaller subset of languages including English, French, Portuguese and others. Nonetheless, language is not a barrier to navigating through the material and identifying the relevant parameters. The involved team members can navigate through reports in such languages. If needed, automatic translation tools were used.

The World Bank's data platform provided GDP share by sector for agriculture, manufacturing, and services. However, GDP share by construction, mining, and energy was also needed to align the data structure with the MAED tool. To address the lack of data available for these sectors, the authors assumed that construction, mining, manufacturing, and energy all fall within the industry sector. Thus, to obtain data for the three remaining sectors, the remaining percentage after considering agriculture, manufacturing, and services from The World Bank's data platform, was divided by three. It is therefore assumed that the GDP share of the construction, mining, and energy sectors are the same.

3. Ethics Statement

Not applicable.

4. CRediT Author Statement

Naomi Tan: Investigation, Conceptualisation, Methodology; Data Collection; Visualization, Writing and Editing; **Robert Ambunda:** Data Collection; Investigation; Writing and Editing; **Nikola Medimorec:** Conceptualisation; Methodology; Data Collection; Investigation; Writing, Review & Editing; Supervision; **Angel Cortez:** Data Collection; **Agustina Krapp:** Data Collection; **Erin Maxwell:** Data Collection; **John Harrison:** Supervision; **Mark Howells:** Supervision

Acknowledgements

We would like to acknowledge the SLOCAT Partnership on Sustainable, Low Carbon Transport who helped make this and future iterations possible. We would also like to acknowledge the International Road Federation (IRF) and the International Union of Railways (UIC) for providing us with these data. The data are extracted from IRF World Road Statistics (WRS) and their use is subject to copyright and specific Terms and Conditions available on the WRS website. More WRS data are available for free on its Data Warehouse www.worldroadstatistics.org. Likewise, data was extracted from the UIC Statistics Rail Information System and Analyses (Railisa) and more can be found on its online tool https://uic-stats.uic.org/

Funding

As well as support in kind provided by the employers of the authors of this note, we also acknowledge core funding from the Climate Compatible Growth Program (#CCG) of the UK's Foreign Development and Commonwealth Office (FCDO). The views expressed in this paper do not necessarily reflect the UK government's official policies.

Declaration of Competing Interests

The authors declare that they have no known competing financial interests or personal relationships which have or could be perceived to have influenced the work reported in this article.